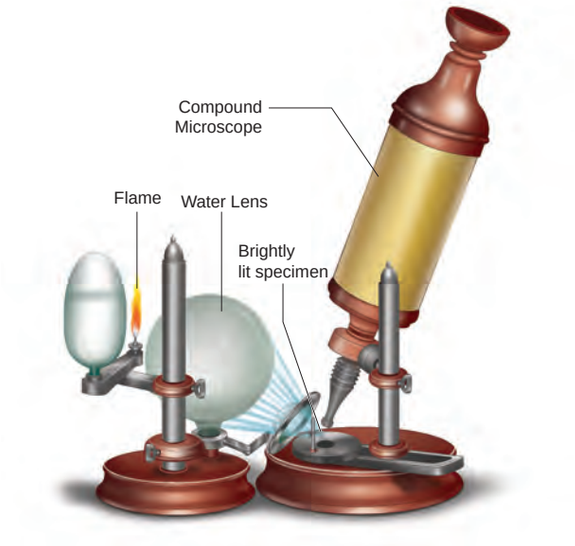
in those days, and built a compound microscope. He placed water-lens beside the microscope to focus the light from an oil-lamp on specimens to illuminate them brightly. So that he was able to see the minute parts of the objects clearly.





Robert Hooke

One day Hooke made thin sections of the cork and observed them through his microscope. He observed many small identical chambers which were hexagonal in shape. He was surprised.

After that he observed many objects like Butterfly's wings, Bee's compound eyes etc.,

Based on this observations, Hooke published a book named Micrographia in the year 1665, where he first used the term Cell. He described the structure of tissue using the term cell.

In Latin the word 'cellua' means a small chamber.

The branch of science that deals with the study of cells is called **'Cell Biology'.**

5.2 The Structural Organization Of The Cell

A typical cell consists of three major parts:

1. An outer **cell membrane.**
2. A liquid **cytoplasm.**
3. A **nucleus.**

Analogous to the body's internal organ, like eyes, heart, lungs, etc..., organelles are specialized structures and perform valuable functions necessary for normal cellular operation. Many distinct structures called **Organelles** lie within the cell.

5.2.1 Size of the cell

The size of cells may vary from a micrometer (a million of a metre) to a few centimeters. Most cells are microscopic and cannot be seen with the naked eye. They can be observed only through the Microscope.

Smallest size of the cell is present in Bacteria. The size of the bacterial cell ranges from 0.01 micrometer to 0.5 micro meter.

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